P2

Force, Motion & Pressure

Physics | Classes 6, 7 & 8 Combined

Classes 6 • 7 • 8 | TET Paper II Science

A · FORCE — Definition, Types & Effects

Force	A **push or pull** by an animate or inanimate agency — causes motion, stops motion, changes direction, or changes shape of an object.
Force	External agency that changes or tends to change the state of rest, uniform motion, direction, or shape of a body. Force is a **vector quantity** (has magnitude AND direction).

SI Unit of Force	**Newton (N)** (CGS unit = dyne | 1 N = 10^5 dyne)
Force Formula	**$F = m \times a$** (Newton's 2nd Law | m in kg, a in m/s^2)
Weight Formula	**$W = m \times g$** ($g = 9.8\ m/s^2 \approx 10\ m/s^2$ at Earth's surface)

Contact & Non-Contact Forces

■ Contact Forces

- Act only when bodies physically touch
- Muscular force — muscles exert push/pull
- Friction — opposes sliding/rolling
- Applied force — push a book, kick a ball
- Tension in a rope, Normal reaction
- Examples: kicking ball, pulling cart, wind on grass

■ Non-Contact Forces

- Act without any physical contact (at a distance)
- Gravity — Earth pulls every object downward
- Magnetic force — magnet attracts iron nail
- Electrostatic force — charged body attracts paper
- Examples: falling coconut, compass needle deflecting
- A magnet attracts iron without touching it

Effects of Force

Effect on Object	Explanation	Real Example
Rest → Motion	Stationary body begins to move when force applied	Kicking a ball at rest
Motion → Rest	Moving body slows down and stops	Braking a bicycle
Change speed	Velocity increases (acceleration) or decreases (deceleration)	Car speeding up / slowing down
Change direction	Direction of moving body changes on applying force	Batsman deflects cricket ball

Effect on Object	Explanation	Real Example
Change shape / size	Object compressed, stretched, bent, or deformed	Squeezing balloon; pulling rubber band
Effect ∝ force magnitude	Greater the force → greater the change in motion	Harder bat hit → ball travels farther

■ Common Misconceptions — Force

■ Force does NOT always produce motion — balanced forces keep object at rest.

■ Weight ($W = mg$) is a FORCE (in Newtons), NOT the same as mass (in kg).

■ A body can be in motion even with ZERO net force — uniform velocity needs no force.

B · MOTION — Rest, Types & Speed

Motion	Change in **position** of an object with respect to time and a reference point (surroundings).
Rest	No change in position with respect to time and surroundings. *Both rest and motion are RELATIVE — not absolute.*

■ REST & MOTION ARE RELATIVE — An object at rest for one observer may be in motion for another.

Types of Motion Based on Path

Type of Motion	Path Description	Class 6 Textbook Examples	More Examples
Linear (Rectilinear)	Straight-line path	Person walking on straight road; falling stone	Car on straight highway
Curvilinear	Curved path (direction changes)	Thrown paper aeroplane	Javelin throw; ball rolled off table
Circular	Circular path around fixed centre	Stone on rope swirled in circles	Earth revolving around Sun; fan blade tip
Rotatory	Spinning about own axis	Spinning top; pencil in sharpener	Earth's rotation; ceiling fan; motor shaft
Oscillatory	To-and-fro about mean position	Pendulum; pencil held between fingers swung	Swing; vibrating guitar string; sewing machine needle
Zigzag (Irregular)	No fixed path or direction	Fly buzzing in room	Brownian motion; people in crowd

■ Periodic Motion	■ Non-Periodic Motion
• Repeats after a fixed time interval (time period) • Oscillatory motion → always periodic • Circular motion → periodic (Earth's revolution: 365 days) • Rotatory motion → periodic (Earth's rotation: 24 hours) • Pendulum swings every fixed interval	• No fixed time pattern — does not repeat regularly • Zigzag / Irregular motion (fly in room) • Curvilinear motion (thrown ball) • Linear motion may be non-periodic • Walking, running in general

Speed — **s = d / t** (Distance ÷ Time | SCALAR | SI unit: m/s)

Speed Type	Definition	Example
Uniform Speed	Equal distances in equal time intervals	Light through vacuum (3×10^8 m/s)
Non-uniform Speed	Unequal distances in equal time intervals	Bus in city traffic
Average Speed	Total distance ÷ total time taken	Overall trip when speed varies

C · Distance, Displacement & Velocity

■ Distance	■ Displacement
• Total length of path from start to end • SCALAR — magnitude only, no direction • SI unit: metre (m) • Always positive (≥ 0) • Can never be zero unless object never moved • Symbol: d	• Shortest straight-line path from start to end • VECTOR — has both magnitude AND direction • SI unit: metre (m) • Can be positive, negative, or ZERO • Zero when object returns to starting point • Symbol: s (or Δx)

■ Distance = Displacement ONLY when object moves in a STRAIGHT LINE from start to end without turning back.

■ Displacement = 0 means the object RETURNED to its starting point (e.g., one complete circular trip).

Nautical Mile	1 nautical mile = 1.852 km — unit for sea/air navigation. **Knot** = 1 nautical mile/hour — speed unit for ships and aircraft.

Speed vs Velocity

	Speed	Velocity
Definition	Rate of change of distance	Rate of change of displacement
Type	Scalar (magnitude only)	Vector (magnitude + direction)
Formula	speed = distance / time	velocity = displacement / time
SI Unit	m/s	m/s
Uniform	Equal distances in equal time	Equal displacement in equal time in same direction
Can be zero?	Only if not moving	Yes — if displacement = 0 in that interval
Example	Car speedometer reading	Car at 60 km/h heading NORTH

Average Speed — **v_avg = Total Distance / Total Time** (Scalar)

Average Velocity — **v_avg = Total Displacement / Total Time** (Vector — direction matters)

Unit Conversion — **1 km/h = 5/18 m/s | 1 m/s = 3.6 km/h** (Multiply km/h by 5/18 to get m/s)

D · Acceleration — All Types

Acceleration	Rate of change of velocity. If a body changes its speed OR direction, it is said to be accelerated. SI unit: m/s^2

Acceleration Formula — **a = (v – u) / t** (v = final velocity, u = initial velocity, t = time | SI: m/s^2)

Type (Section)	Definition	Sign	Textbook Example
Positive Acceleration (2.3.1)	Velocity INCREASES with time	+ve	Car at rest reaches 12 m/s in 4 s → a = 3 m/s^2
Negative Acceleration / Deceleration / Retardation (2.3.2)	Velocity DECREASES with time	–ve	Golf ball: 8 m/s → 2 m/s in 10 s → a = –0.6 m/s^2
Uniform Acceleration (2.3.3)	Change in velocity is SAME each second	±	Bus velocity: 20, 40, 60, 80… m/s — increases by 20 m/s every second
Non-uniform Acceleration (2.3.4)	Change in velocity is DIFFERENT each second	±	Velocity sequence: 0, 10, 40, 60, 70, 50 m/s — change varies

■ Did You Know? (Textbook Science Facts)

◆ Cheetah accelerates from 0 to 20 m/s in just 2 seconds — acceleration = 10 m/s^2 (Class 7 textbook).

◆ Usain Bolt runs 100 m in 9.58 s → average speed ≈ 10.44 m/s (discussed in Class 7).

◆ Tortoise speed: 0.1 m/s | Person walking: 1.4 m/s | Falling raindrop: 9–10 m/s (Class 7 textbook data).

◆ Badminton smash speed: 80–90 m/s | Rocket: 5200 m/s | Passenger jet: 180 m/s.

E · Distance–Time & Speed–Time Graphs

Distance–Time Graph — Section 2.4 (4 Cases)

Case	Graph Shape	Gradient / Slope	Motion Represented
1	Horizontal line	Zero (flat)	Object at REST — distance does not change
2	Straight line rising	Constant (positive)	Uniform speed — equal distance per second
3	Curve — concave (bending upward)	Increasing	Speed INCREASING — acceleration
4	Curve — convex (flattening)	Decreasing	Speed DECREASING — deceleration

Speed–Time Graph — Section 2.5 (6 Cases)

Case	Graph Shape	What it Means
1	Horizontal line at speed = 0	Object at rest (zero speed, zero acceleration)
2	Horizontal line at speed > 0	Uniform speed — zero acceleration
3	Straight line RISING (constant slope)	Uniform acceleration — speed increases at constant rate
4	Straight line FALLING (constant slope)	Uniform deceleration — speed decreases at constant rate
5	Rising CURVE with increasing slope	Non-uniform acceleration — acceleration itself is increasing
6	Rising CURVE with decreasing slope	Non-uniform acceleration — acceleration is decreasing (speed still rising but slower)

D–T vs S–T Graph Comparison — Section 2.5.1

Journey Phase	Distance–Time Graph	Speed–Time Graph
Accelerating uniformly from rest	Concave curve (gradient increases)	Straight line rising (positive constant slope)

Journey Phase	Distance–Time Graph	Speed–Time Graph
Moving at constant speed	Straight line (constant gradient)	Horizontal line (slope = 0 = zero acceleration)
Decelerating uniformly to stop	Convex curve (gradient decreases)	Straight line falling (negative constant slope)
Important: Area under graph	Not directly useful	= Distance travelled (area of triangle/rectangle)

■ Graph Reading Tips

- ■ D–T and S–T graphs may look similar — always check the Y-axis label first.
- ■ Slope of D–T graph = speed | Slope of S–T graph = acceleration.
- ■ Area under S–T graph = distance travelled (calculate as triangle / trapezium area).

F · Centre of Gravity & Stability

Centre of Gravity (CG): The point through which the **entire weight of an object appears to act**. For regular shapes, CG lies at the geometric centre.

CG of Regular-Shaped Objects — Section 2.6.1

Shape	Location of Centre of Gravity	How to find for irregular shapes
Square / Rectangle	Intersection of diagonals	—
Circle / Disc	Centre of the circle	—
Triangle	Centroid — where three medians intersect	—
Ring / Hollow circle	Centre (may be in empty space)	—
Uniform ruler / rod	Midpoint of the ruler	—
Irregular lamina	Found experimentally	Suspend from 3 holes; draw plumb-line each time; intersection = CG

Types of Equilibrium / Stability — Section 2.7

Type	When Displaced...	CG change	Returns?	Example
Stable Equilibrium	CG rises; weight-line stays WITHIN base area	**Rises ↑**	**YES ✓**	Frustum on wide base; Thanjavur doll; racing car

Unstable Equilibrium	CG lowers; weight-line falls OUTSIDE base area	**Falls ↓**	**NO ✗**	Frustum balanced on tip; pencil on tip
Neutral Equilibrium	CG remains at SAME height; body rolls without toppling	**No change**	**Stays in new pos.**	Frustum on its curved side; sphere on flat surface

Conditions & Applications of Stability — Section 2.7.1 & 2.7.2

Stability INCREASES by:

1. **Lowering the Centre of Gravity**

→ Use a heavy base (base absorbs weight)

2. **Increasing the base area**

→ Wider base = more stable

Thanjavur Doll:

- Traditional Thanjavur terracotta toy
- CG concentrated at **bottommost point**
- Returns to upright on any tilt → stable equilibrium
- Shows slow oscillatory dance-like motion

Real-Life Applications:

- Tour bus: luggage stored at **BOTTOM** (not on roof)
- Double-decker bus: extra passengers NOT on upper deck
- Racing cars: built **low and broad**
- Table lamps / fans: **large heavy base**

Finding CG of irregular shape:

Step 1 → Make 3 holes in the lamina

Step 2 → Suspend from each hole + hang plumb-line

Step 3 → Draw lines along plumb-line on lamina

Step 4 → Intersection of 3 lines = Centre of Gravity

G · Pressure — Thrust, Atmospheric & Liquid

Thrust	Force acting **perpendicularly** on a given surface area. Unit: Newton (N). A given force on a smaller area produces greater thrust effect.
Pressure	Amount of thrust (force) acting on **unit area** of surface. SI unit: **Pascal (Pa) = 1 Nm^{-2}** — named after French scientist Blaise Pascal.
Pressure Formula	**$P = F / A$ = Thrust / Area** (1 Pascal = 1 Nm^{-2} | CGS: $dyne/cm^2$)

Pressure — Effect of Area on Practical Situations

Situation	Force	Area	Pressure Effect	Reason
Sharp knife / nail	Same	Very small (tip)	Very HIGH	Small area → concentrates force → penetrates easily
Injection needle	Same	Tiny tip	Very HIGH	Concentrated force → pierces skin without large force
Camel's padded feet	Body weight	Large	LOW	Large feet area spreads weight → no sinking in sand

Situation	Force	Area	Pressure Effect	Reason
Heavy truck wheels	Body weight	Many tyres (large total)	LOW	Many wheels increase contact area → less road pressure
Broad bag strap	Bag weight	Wide strap	LOW	Wide strap spreads load → comfortable on shoulder
Dam wall — wider base	Water	Larger at bottom	Withstands high pressure	Liquid pressure increases with depth → thicker wall needed at base

Atmospheric Pressure

Atmospheric Pressure	Weight of the column of atmospheric air above **unit area** of Earth's surface. It acts in ALL directions — not just downward.

Parameter	Value / Explanation
Standard value (sea level)	1 atm = 76 cm Hg = 760 mmHg = 1,01,325 Pa ≈ 1.01×10^5 Nm^{-2}
Measuring instrument	Barometer (mercury barometer) — invented by Evangelista Torricelli (1643)
Pressure with altitude	Atmospheric pressure DECREASES as altitude increases above Earth's surface
Tilting the barometer tube	Mercury level remains the same — proof that air pressure acts equally in all directions
Aneroid barometer	No liquid — uses elastic metal box deformation; used in aircraft as altimeter
Cooking at high altitude	Lower atmospheric pressure → boiling point of water drops (even 80°C) → food does not cook properly
Pressure cooker	Sealed vessel → increased internal pressure → boiling point of water rises → food cooks faster

Liquid Pressure

Liquid Pressure	$P = \rho \times g \times h$ (ρ = density (kg/m^3), g = 9.8 m/s^2, h = depth below surface (m))

Property	Explanation	Textbook Experiment / Example
Increases with depth	Greater depth → more liquid weight above → more pressure	Spouting can: lowest hole shoots water farthest
Same in ALL directions at a given depth	Liquid exerts equal pressure sideways, upward, downward at same depth	Rubber ball with holes: water shoots equally from all holes

Property	Explanation	Textbook Experiment / Example
Depends only on height of liquid column	Not on shape/width of container — only vertical height (h) matters	Balloon at bottom of tube bulges more as water height increases
Buoyant Force (Archimedes)	Upward force on submerged body = weight of liquid displaced	Ship floats; submarine; hydrometer measures density
Object floats if...	Weight of object < Buoyant force (upward force)	Cork floats in water
Object sinks if...	Weight of object > Buoyant force	Iron ball sinks in water
Scuba diver suits	Withstand enormous water pressure at great ocean depths	Pressure increases with depth — special suit prevents body compression

Pascal's Law

Pascal's Law: Pressure applied at any point of a **liquid at rest in a closed container** is transmitted equally in all directions throughout the liquid. (Blaise Pascal, 1648)

1 Apply small force on small piston	**2** Pressure = F■/A■ created	**3** Pressure transmitted equally to ALL points	**4** Same pressure acts on large piston (A■)	**5** Large force F■ = P × A■ produced

Application	Working Principle	Where Used
Hydraulic Lift / Jack	F■/A■ = F■/A■ → small force on small piston → large force lifts vehicle	Car service stations; aircraft jacks
Hydraulic Brakes	Brake fluid transmits force equally to all four wheel cylinders simultaneously	Cars, buses, motorcycles
Hydraulic Press	Large mechanical advantage → compress materials	Cotton/paper bale compression
Syringe / Pump	Pressure on liquid transmitted uniformly to needle outlet	Medical injections; pumps

H · Surface Tension & Viscosity

Surface Tension: Property of liquid surface molecules to contract and minimize surface area. Defined as **force per unit length** acting on the surface. Unit: **Nm^{-1}**.

Viscosity / Viscous Force: Frictional force between **successive moving layers** of a liquid that opposes their relative motion. CGS unit: **Poise**. SI unit: **$kg\ m^{-1}\ s^{-1} = N\ s\ m^{-2}$**.

■ Surface Tension	■ Viscosity (Viscous Force)
• Acts at the SURFACE / boundary of liquid • Contracts surface to MINIMUM area • Unit: Nm^{-1} (Newton per metre) • Decreases when temperature rises • Decreased by detergent/soap • Causes: spherical drops, capillary rise, insects on water	• Acts BETWEEN LAYERS of a moving liquid • Opposes relative motion of layers (internal friction) • CGS: poise | SI: kg m^{-1} s^{-1} • Also decreases when temperature rises • Honey > Oil > Water — more viscous = flows slower • Causes: slow flow of thick liquids; engine oil lubricates

Applications of Surface Tension — Section 2.5.1

Surface Tension Applications:

- **Rain drops spherical** — liquid contracts to minimum surface area for given volume
- **Water strider insect** walks on water without sinking
- **Paper clip / needle floats** — though denser than water
- **Capillary action in plants** — water rises in narrow xylem vessels against gravity
- **Soap bubble** formation — liquid film holds shape due to surface tension

Practical Consequences:

- **Detergent reduces surface tension** — water spreads into cloth fibres → better cleaning
- **Sailors pour oil on rough sea** — reduces surface tension → calms waves and protects ship
- **Water spreads on glass** (wets glass) because adhesion > cohesion
- **Mercury beads up on glass** (does not wet) because cohesion > adhesion
- **Mosquito larvae** breathe through water surface — surface tension supports them

■ Did You Know? (Textbook Science Facts)

◆ Surface tension of water = 0.073 N/m at 20°C — pure water has higher surface tension than soapy water.

◆ Water strider insect has waxy legs — increases contact angle, prevents wetting, allows walking on water surface.

◆ Blood viscosity is about 3–4 times that of water — important for cardiovascular health.

◆ Viscosity of honey: ~2000–10,000 times that of water — why it flows very slowly at room temperature.

I · Friction — Types, Factors & Control

Friction	Force arising between two bodies in contact that **opposes relative motion**. Caused by geometrical dissimilarities (roughness/irregularities) on surfaces. Always acts **opposite to direction of motion**.

Types of Friction — Section 2.7.1

Type	When it Acts	Magnitude Order	Class 8 Example
Static Friction	Body at REST; external force applied but no motion yet	LARGEST	Book resting on table; parked car; wall before pushing
Sliding Friction (Kinetic)	During SLIDING of one surface over another	MODERATE	Box dragged on floor; chalk on blackboard; braking
Rolling Friction	When body ROLLS over another surface	SMALLEST	Ball rolling on ground; bicycle/car tyre rolling on road
Fluid Friction (Viscosity)	Object moving through liquid or gas (air/water resistance)	Varies	Ship in water; aeroplane in air; swimming

■ **Order of friction magnitude: Rolling < Sliding < Static (rolling friction is smallest → ball bearings used in machines)**

Factors Affecting Friction — Section 2.7.2

Factor	Effect	Textbook Example
Nature of surface (rough vs smooth)	Rough surface → MORE friction; Smooth/polished → LESS friction	Walking on wet surface is slippery; polished marble floor more slippery than rough road
Weight / Mass of body	Greater mass/weight → GREATER frictional force	Loaded trolley harder to push than empty trolley; cyclist with heavy load pedals harder
Area of contact	Greater contact area → greater friction (for same weight)	Road roller (broad drum) offers more friction; cycle tyre (narrow) offers less

Advantages vs Disadvantages of Friction — Sections 2.7.3 & 2.7.4

■ Advantages of Friction	■ Disadvantages of Friction
• We walk without slipping on ground • Vehicles brake and stop safely • We write with pen/pencil on paper • Knots, nails, bolts, screws hold firmly • Matchstick lights when struck on box • Climbing, gripping, stitching clothes possible • Brakes on cycle/car work due to friction	• Wears out surfaces — shoe soles, tyres, gear teeth • Wastes energy — extra effort to overcome friction • Produces HEAT — damages machines and bearings • Moving parts make noise due to friction • Vehicles slow down due to air and road friction • Engine parts wear out → frequent maintenance needed • Reduces efficiency of all machines

Friction is a "Necessary Evil"	Essential for daily life (walking, braking, writing) yet also destructive (wears surfaces, wastes energy). Hence called a **necessary evil**.

Increasing & Decreasing Friction — Section 2.7.5

Method	Effect on Friction	How it Works	Example
Increase area of contact	INCREASES	More contact area → more surface irregularities engaged	Brake shoes pressed close to wheel rim
Lubricants (oil/grease)	DECREASES	Fills gaps in rough surfaces; smooth layer prevents direct contact	Engine oil, grease, coconut oil, castor oil, graphite
Ball Bearings	DECREASES	Converts sliding → rolling friction (rolling much smaller)	Cycle hub, electric motors, axles, wheels
Streamlining body shape	DECREASES fluid friction	Smooth shape reduces air/water resistance	Aircraft, ships, sports cars, fish body
Polishing / smoothing surfaces	DECREASES	Removes surface irregularities → smoother contact	Sports tracks, machine parts, bearings
Using rough grooved surface	INCREASES	More grip needed for safety	Tyre treads, brake pads, shoe soles, stair nosing

J · Solved Numerical Problems

Problem 1	A vehicle covers 400 km in 5 hours. Find speed.
Given	d = 400 km = 4,00,000 m; t = 5 h = 18,000 s

Solution	Speed = d/t = 400/5 = 80 km/h OR = 4,00,000/18,000 = 22.22 m/s
Answer	**80 km/h = 22.22 m/s**

Problem 2	A car starts from rest and reaches 20 m/s in 10 s. Find acceleration.
Given	u = 0 m/s, v = 20 m/s, t = 10 s
Solution	a = (v–u)/t = (20–0)/10 = 2 m/s^2
Answer	**Acceleration = 2 m/s^2**

Problem 3	Golf ball decelerates from 8 m/s to 2 m/s in 10 s. Find deceleration.
Given	u = 8 m/s, v = 2 m/s, t = 10 s
Solution	a = (v–u)/t = (2–8)/10 = –6/10 = –0.6 m/s^2
Answer	**Deceleration = 0.6 m/s^2**

Problem 4	Elephant weight = 4000 N, one foot area = 0.1 m^2. Find pressure on one foot.
Given	Force on one foot = 4000 ÷ 4 = 1000 N; A = 0.1 m^2
Solution	P = F/A = 1000/0.1 = 10,000 Nm^{-2}
Answer	**10,000 Pa = 10^4 Nm^{-2}**

Problem 5	Stone weighs 500 N, contact area = 25 cm^2. Find pressure.
Given	F = 500 N; A = 25 cm^2 = 25×10^{-4} m^2 = 0.0025 m^2
Solution	P = F/A = 500/0.0025 = 2,00,000 Nm^{-2}
Answer	**2 × 10^5 Pa**

Problem 6	Geetha cycles at 2 m/s for 15 min. Find distance to school.
Given	v = 2 m/s; t = 15 min = 15×60 = 900 s
Solution	Distance = v × t = 2 × 900 = 1800 m
Answer	**1800 m = 1.8 km**

K · Key Facts — Rapid Revision (Numbers, Years & Names only)

Formulas, Values & Units

Quantity / Concept	Formula / Value	Unit / Note
Force	F = m × a	Newton (N) | CGS: dyne
1 Newton	= 10^5 dyne	Unit conversion

Quantity / Concept	Formula / Value	Unit / Note
Weight	$W = m \times g$ ($g = 9.8\ m/s^2 \approx 10$)	Newton (N)
Speed	distance ÷ time	m/s (scalar)
Velocity	displacement ÷ time	m/s (vector)
Acceleration	$a = (v – u) / t$	m/s^2 (vector)
1 km/h	= 5/18 m/s ≈ 0.278 m/s	Unit conversion
1 m/s	= 3.6 km/h (multiply by 3.6)	Unit conversion
Pressure	P = F/A = Thrust ÷ Area	Pascal (Pa) = Nm^{-2}
1 atmosphere	= 76 cm Hg = 760 mmHg = 1.01×10^5 Pa	Standard sea-level value
Liquid pressure	$P = \rho \times g \times h$	Pa (Nm^{-2})
Surface tension unit	Nm^{-1} (Newton per metre)	Force per unit length on liquid surface
Viscosity (CGS)	Poise	CGS unit
Viscosity (SI)	$kg\ m^{-1}\ s^{-1} = N\ s\ m^{-2}$	SI unit
1 Nautical mile	= 1.852 km	Sea/air navigation; 1 knot = 1 nmi/h
Cheetah speed	25–30 m/s | 0→20 m/s in 2 s = 10 m/s^2 acc.	Class 7 textbook data
Speed of light	3×10^8 m/s (vacuum)	Uniform velocity example
Tortoise speed	0.1 m/s | Person walking: 1.4 m/s	Class 7 textbook data
Friction order	Rolling < Sliding < Static	Smallest to largest friction

Scientists, Discoverers & Years

Scientist	Contribution	Year / Class
Isaac Newton	Laws of Motion; Law of Gravitation; unit "Newton" named after him	1687
Blaise Pascal	Pascal's Law of fluid pressure; unit "pascal" named after him	1648

Scientist	Contribution	Year / Class
Evangelista Torricelli	Invented the mercury barometer to measure atmospheric pressure	1643
Galileo Galilei	Law of falling bodies; studied motion, velocity, acceleration	1638
Archimedes	Archimedes' Principle: buoyant force = weight of fluid displaced	287–212 BC
Aryabhata	Observed rest and motion are RELATIVE (star/river bank example)	5th century CE

L · Glossary — Definitions (one-line definitions only)

Force — Push or pull by animate/inanimate agency; vector quantity; SI unit: Newton	**Thrust** — Force acting perpendicularly on a surface area; unit: Newton
Pressure — Thrust per unit area (P = F/A); SI unit: Pascal (Pa)	**Newton (N)** — SI unit of force; 1 N = force that gives 1 kg an acceleration of 1 m/s^2
Pascal (Pa) — SI unit of pressure; 1 Pa = 1 Nm^{-2}; named after Blaise Pascal	**Motion** — Change in position of a body with respect to time and a reference point
Rest — No change in position with respect to time and a reference point	**Distance** — Total length of path travelled; scalar quantity; unit: metre
Displacement — Shortest straight-line path from initial to final position with direction; vector	**Speed** — Rate of change of distance; scalar; unit: m/s
Velocity — Rate of change of displacement; vector; unit: m/s	**Acceleration** — Rate of change of velocity; a = (v–u)/t; unit: m/s^2
Deceleration — Negative acceleration; velocity decreasing with time; also called retardation	**Linear Motion** — Motion along a straight-line path
Circular Motion — Motion along a circular path around a fixed centre or axis	**Rotatory Motion** — Spinning of a body about its own axis
Oscillatory Motion — Repeated to-and-fro motion about a mean (equilibrium) position	**Curvilinear Motion** — Motion along a curved path where direction continuously changes
Periodic Motion — Motion that repeats identically after a fixed time interval (time period)	**Uniform Motion** — Motion with equal distances covered in equal time intervals
Average Speed — Total distance divided by total time taken for a journey	**Average Velocity** — Total displacement divided by total time taken for a journey
Atmospheric Pressure — Weight of atmospheric air acting per unit area of Earth's surface	**Barometer** — Instrument to measure atmospheric pressure; mercury barometer invented by Torricelli
Buoyant Force — Upward force exerted by a fluid on an immersed or floating object	**Pascal's Law** — Pressure applied to a confined liquid is transmitted equally in all directions

Hydraulic Machine — Device using Pascal's Law to multiply force via liquid pressure transmission	**Surface Tension** — Property of liquid surfaces to contract to minimum area; unit: Nm^{-1}
Viscosity — Property of a liquid to resist flow; internal friction between moving liquid layers	**Viscous Force** — Frictional force between adjacent layers of a liquid moving at different velocities
Friction — Force opposing relative motion between surfaces in contact; due to surface irregularities	**Static Friction** — Friction on a body at rest; attains maximum value just before motion begins
Sliding Friction — Kinetic friction during sliding motion of one surface over another	**Rolling Friction** — Friction when a body rolls over a surface; less than sliding friction
Lubricant — Substance that reduces friction by forming smooth layer between surfaces (oil, grease)	**Centre of Gravity** — Point through which total weight of an object appears to act
Stable Equilibrium — Object returns to original position when slightly displaced; CG rises on displacement	**Unstable Equilibrium** — Object topples further when displaced; CG lowers on displacement
Neutral Equilibrium — Object stays in displaced position; CG height unchanged on displacement	**Capillarity** — Rise or fall of liquid in a narrow tube due to surface tension and adhesion
Knot — Unit of speed: 1 knot = 1 nautical mile per hour; used in sea and air navigation	